

In Silico Design of High-Affinity gp100 Antigenic Mimotopes for Targeting Melanoma

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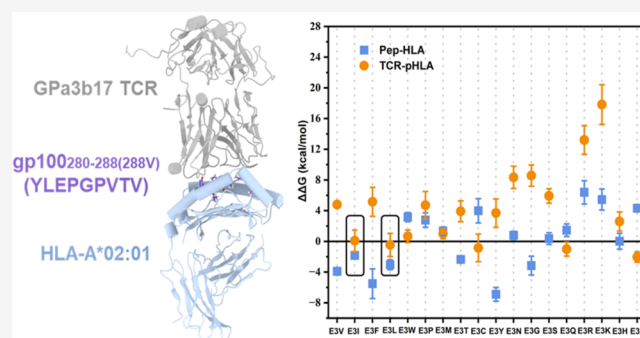


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ABSTRACT: Human CD8⁺ cytotoxic T lymphocytes (CTLs) induce melanoma regression by utilizing T-cell receptors (TCRs) to recognize tumor-specific epitopes. One such epitope, gp100_{280–288(288 V)} (YLEPGPVTV), is presented by human leukocyte antigen (HLA)-A*02:01 molecules. Computational approaches, particularly the predictions on peptide–HLA (pHLA) and TCR–pHLA binding affinities, are complementary to experimental approaches in guiding further optimization and design of the gp100_{280–288(288 V)} peptide (mimotopes) for enhanced antitumor responses. Here, with all-atom molecular dynamics (MD) simulations and free energy perturbation (FEP) methods, we designed a series of potential mimotopes through *in silico* mutagenesis studies of gp100_{280–288(288 V)} utilizing the latest affinity-matured TCR structure. Our results showed that E3I and E3L mutants increased HLA binding without affecting TCR binding, indicating their potential as epitope candidates. Conversely, the E3A and E3V mutations improved peptide–HLA binding but weakened the TCR–pHLA binding affinity. In-depth structural analysis revealed that insufficient occupancy of the HLA pocket by residue 3 in the E3A and E3V mutants would induce a twisted peptide conformation at positions 4 to 6, disrupting the interaction between TCR and pHLA. Therefore, it is crucial to consider the impact of single-residue mutations that alter peptide conformation on TCR binding when designing effective antigenic mimotopes. Furthermore, the E3I+V7L and E3L+V7L double mutants are also found to produce favorable binding affinity changes similar to those of the E3I and E3L mutants, suggesting their potential as advantageous mimotope candidates. Our work provides structural insights for the rational design of favorable gp100 mimotopes and may contribute to developing gp100 epitope-based vaccines.



INTRODUCTION

Cytotoxic T lymphocytes (CTLs) initiate a targeted immune response against melanoma cells by recognizing tumor-derived peptides presented on the cell surface by human leukocyte antigens (HLAs).¹ For example, epitopes from melanocyte differentiation proteins can be identified as shared melanoma-associated epitopes by T-cell receptors (TCRs) on the patient's lymphocytes.^{2,3} Glycoprotein 100 (gp100), a 661-amino-acid protein, has emerged as a promising target in melanoma immunotherapy.^{4,5} This protein plays a crucial role in melanosome development and melanin production.⁵ Notably, gp100 is often overexpressed in melanoma cells during all stages of tumor progression compared to normal melanocytes.^{6,7}

Previous studies have demonstrated that epitopes derived from gp100 are recognized by tumor-infiltrating lymphocytes and circulating T-cells, leading to tumor regression in metastatic melanoma patients undergoing adoptive cell therapy.^{1,8–10} Among these epitopes, the nonamer peptide gp100_{280–288(288A)} (YLEPGPVTA) was recognized by HLA-A*02:01-restricted tumor-infiltrating lymphocytes in melanoma patients.¹¹ Parkhurst et al. demonstrated that this peptide

could induce CTL responses *in vitro*, suggesting its immunogenicity.¹² Furthermore, gp100_{280–288(288A)}, naturally processed and presented at relatively low levels of the HLA-A*02:01 molecule, is sufficient for CTL recognition and subsequent tumor lysis.¹³ Thus, these findings suggest that gp100_{280–288(288A)} is a promising epitope for melanoma therapeutics. Currently, several mimotope candidates have been developed by further optimizing the gp100_{280–288(288A)} peptide.^{12,14,15} Mimotopes are short peptides that mimic natural antigenic epitopes, including complex and discontinuous ones. They can replace native epitopes and induce similar immune responses, including antibody production, when used in vaccines.¹⁶ Among these, a modified version of gp100_{280–288(288A)}, where the alanine at anchor position 9 is

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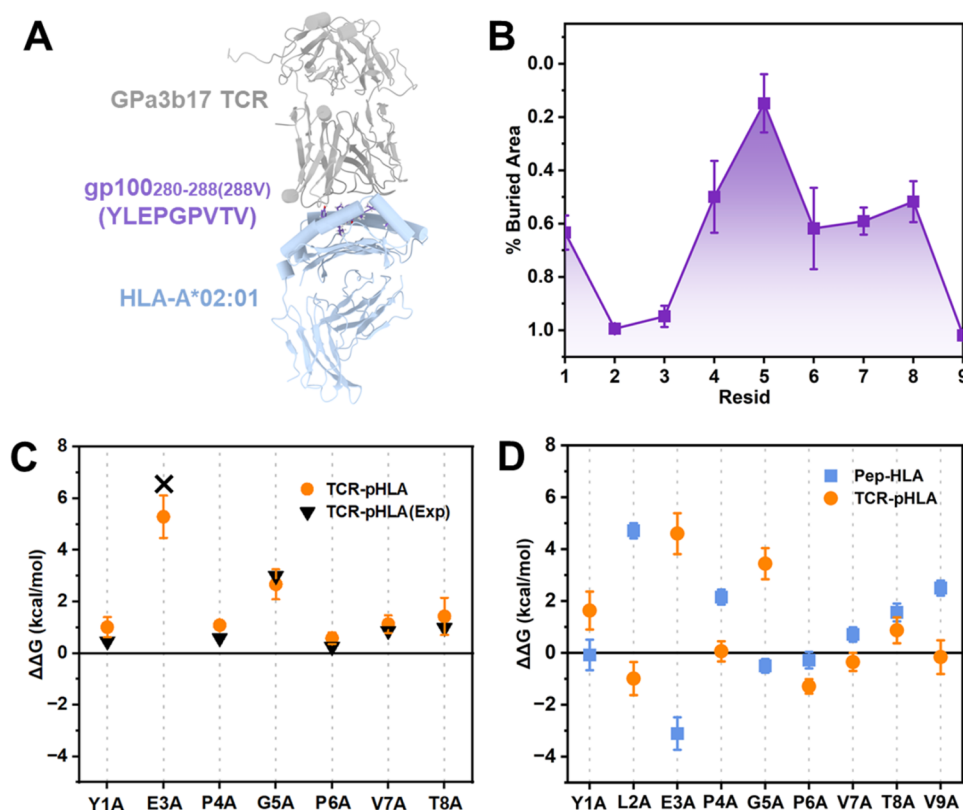


Figure 1. (A) Crystal structure of the GP3b17 TCR-gp100_{280-288(288V)} peptide-HLA-A*02:01 (GP3b17 TCR-pHLA) complex (PDB ID: 7PHR). The gp100_{280-288(288V)} peptide is shown in a purple stick representation. The GP3b17 TCR and HLA-A*02:01 molecules are shown in gray and blue cartoon representations, respectively. (B) The buried surface area of the gp100_{280-288(288V)} peptide in the GP3b17 TCR-pHLA complex. (C) Comparison of relative binding free energy changes $\Delta\Delta G$ from FEP calculations (orange circles) and experimental measurements (black triangles) between PMEL17 TCR and pHLA upon alanine scanning mutations. "X" represents no binding in the experiment (i.e., a large positive $\Delta\Delta G$). The experimental values are derived from previous work.²⁰ (D) Binding free energy changes for peptide-HLA and GP3b17 TCR-pHLA upon alanine scanning mutations.

substituted by valine (gp100_{280-288(288V)}, YLEPGPVTV), has been shown to enhance pHLA stability¹⁷ and significantly increase its immunogenicity.¹⁸ Recently, an affinity-matured TCR- $\alpha\beta$ (GP3b17) has been developed for immunotherapy, namely, ImmTACs, which binds to the gp100_{280-288(288V)} peptide presented by the HLA-A*02:01 molecule. The crystal structure of this ternary complex (GP3b17 TCR-pHLA) has been reported.¹⁹ However, the structure-property relationships and structural dynamics of this effective complex remain largely elusive, which hinders further efforts in optimizing immunogenic mimotopes.

In this study, we conducted a series of computational analyses to explore the structural dynamics and binding properties of the gp100_{280-288(288V)} peptide within the GP3b17 TCR-pHLA complex. Through MD simulations and free energy perturbation (FEP) calculations, our results demonstrated the critical role of the peptide conformation in influencing its binding affinity to HLA and modulating the TCR engagement. Based on these insights, we rationally designed a series of potential mimotope candidates with an enhanced binding affinity for melanoma, which may help guide the development of gp100 epitope-based vaccines.

SYSTEM AND METHODS

Molecular Dynamics Simulation. The complex structure of the GP3b17 TCR binding with the gp100_{280-288(288V)} peptide (YLEPGPVTV) and HLA-A*02:01, namely, GP3b17

TCR-pHLA (Figure 1A), was obtained from the crystal structures deposited in the Protein Data Bank (PDB ID: 7PHR¹⁹). Additionally, the complex of PMEL17 TCR (TRAV21 TRBV7-3) binding with gp100_{280-288(288A)} peptide (YLEPGPVTA) and HLA-A*02:01, i.e., PMEL17 TCR-pHLA (Figure S1, PDB ID: 5EU6²⁰), was used as a reference model to validate the results obtained from FEP calculations against experimental data.²⁰ The experimental relative binding free energy changes caused by mutations can be derived as $\Delta\Delta G_{\text{exp}} = -k_B T \ln(K_d^{\text{wt}}/K_d^{\text{mut}})$, where K_d^{wt} and K_d^{mut} are dissociation constants for WT and mutated gp100 peptide elements, respectively, and k_B is the Boltzmann constant and $T = 310$ K.

All-atom MD simulations were performed using GROMACS 5.1.4²¹ with the CHARMM36 force field.²²⁻²⁴ The CHARMM36 force field is widely used for simulations of biomacromolecular systems due to its validated accuracy, experimental consistency, and good compatibility with simulators such as GROMACS and NAMD.^{25,26} Each system was solvated in a cubic box of TIP3P²⁷ water molecules (15*15*15 nm³). Na⁺ and Cl⁻ ions were added to neutralize the system to a physiological concentration (0.15 M). The particle mesh Ewald method²⁸ was used to calculate the long-range electrostatic interactions. A typical 12 Å cutoff distance was used for van der Waals and short-range electrostatic interactions. The temperature $T = 310$ K was controlled using the velocity-rescaled Berendsen thermostat.²⁹ The pressure

was set to 1 atm using an isotropic Parrinello–Rahman barostat.³⁰ The bond length involving hydrogen atoms was constrained by the LINCS algorithm,³¹ which allows a time step of 2 fs. Each solvated system was energy minimized for 20,000 steps and equilibrated for 10 ns with position restraints on all heavy atoms. The end structures of the equilibrium runs were selected as the starting points for production simulations in the NPT ensemble, with all restraints removed. Three independent simulations were run for the wild-type (WT) and single mutant systems, and two independent simulations for the double mutant system. A summary of the simulations performed in this study is provided in Table S1. The snapshots were rendered by the ChimeraX program.³²

Free-Energy Perturbation (FEP) Calculation. The FEP calculations have been widely used for *in silico* mutagenesis studies, including TCR–pHLA complex binding involved in cancer, T1D, and HIV.^{33–38} Although FEP calculations are more time-consuming compared to the docking score and MM/PBSA methods, they can provide very accurate estimates of binding free energy changes caused by point mutations and are in good agreement with experimental measurements.^{36,39} The binding free energy change, ΔG , of mutation can be calculated as follows

$$\Delta G_{\lambda} = -k_B T \ln \langle \exp(-\beta[V(\lambda + \Delta\lambda) - V(\lambda)]) \rangle_{\lambda} \quad (1)$$

$$\Delta G = \sum_{\lambda} \Delta G_{\lambda} \quad (2)$$

where $V(\lambda) = (1 - \lambda)V_1 + \lambda V_2$, with V_1 and V_2 representing the potential energy of the wild type and the mutant, respectively. The parameter λ changes from 0 to 1 as the system transforms from the wild type to the mutant. Multiple independent FEP calculations were carried out using a starting structure chosen randomly from the corresponding MD production run. Following similar protocols used previously for various TCR–pHLA complex bindings,^{33–37} we adopted a 38 windows strategy with soft-core potential ($\lambda = 0.00, 0.00001, 0.0001, 0.001, 0.01, 0.02, 0.06, 0.1, 0.12, 0.13, 0.14, 0.18, 0.22, 0.26, 0.30, 0.34, 0.38, 0.42, 0.46, 0.5, 0.54, 0.58, 0.62, 0.66, 0.7, 0.74, 0.78, 0.82, 0.86, 0.9, 0.94, 0.98, 0.99, 0.994, 0.999, 0.9994, 0.9999, 0.99999, 1.00$). According to our long-term experience, using denser λ windows near the two ends combined with the soft-core potentials can ensure good convergence in the end point regions, especially for mutations of charged or polar groups. Therefore, when computing resources are sufficient, we prefer to employ the settings described above to ensure more accurate and robust free energy estimates. Quantifying the absolute binding affinity directly is challenging due to the lengthy period of the binding process between two interacting surfaces, such as HLA-A*02:01 and the peptide. To resolve this issue, as illustrated in Figure S2, we estimated the relative binding free energy change $\Delta\Delta G$ using a thermodynamic cycle. Instead of calculating direct binding energies ΔG_B and ΔG_A , we obtained the free-energy changes for the identical mutation in both the bound state (pHLA, ΔG_2) and the free state (peptide, ΔG_1). The overall change in free energy should be zero according to this thermodynamic cycle, and the relative binding affinity for the mutation from A to B is thus:

$$\Delta\Delta G = \Delta G_B - \Delta G_A = \Delta G_2 - \Delta G_1 \quad (3)$$

Five replicas were performed for each mutant, and each replica took 15.2 ns (0.4 ns for each window), giving a total

simulation time of 152 ns (15.2 ns $\times$ 5 replicas $\times$ 2 states). Free energy components for the bound state and the free state were averaged over all five replicas. FEP calculations were performed using NAMD²¹ with the CHARMM36 force field and TIP3P water molecules. The final free energy is calculated by applying exponential averaging (Zwanzig equation) to the potential energy differences from each simulation window and summing the resulting free energy increments across all windows.⁴⁰ Free energy components of the bound and free states were averaged over all five replicas, and the standard errors were considered as the error bars.

Analysis Methods. The VMD “measure SASA” command was used to calculate the buried surface area with a probe radius of 1.4 Å. The buried surface area fraction was calculated as the buried surface area of the peptide residue in the peptide–HLA (pHLA) complex divided by the surface area of the isolated residue.⁴¹ There are other analysis tools available that can also calculate the buried surface area, such as “gmX sasa” in GROMACS, and their results are consistent (Figure S3).

RESULTS

Binding of gp100_{280–288(288 V)} Peptide to HLA-A*02:01 and GPa3b17 TCR. To obtain the structural and dynamic characteristics of the GPa3b17 TCR–pHLA complex (Figure 1A), we first performed multiple independent 700 ns MD simulations of the ternary complex. The stability of the complex was illustrated by the stable root-mean-square deviations (RMSDs) (average RMSD $\sim$ 3.5 Å, see Figure S4) during the course of MD simulations, which also indicates that the use of the CHARMM36 force field in our study is appropriate. Figure 1B illustrates the buried surface area (see Analysis Methods) of the gp100_{280–288(288 V)} peptide inside the peptide-binding groove of HLA. The results indicated that the midregion of the peptide (position 4 to 6) is more exposed and inclined to interact with TCR, and therefore holds physiological relevance for TCR recognition.⁴¹ In contrast, residues near both terminal regions (positions 2 and 9) were identified to be the anchor residues⁴² with high ratios of buried surface area, which should play a vital role in securing the peptide into the HLA binding groove. Moreover, the residue at position 3 exhibits an overall buried percentage of >90%, suggesting a high propensity to interact with the HLA molecule.

In Silico Alanine Scanning of gp100_{280–288(288 V)} Peptide. To assess the contribution of each residue in the gp100_{280–288(288 V)} peptide to peptide–HLA and TCR–pHLA binding, we employed FEP calculations to perform the peptide *in silico* alanine scanning experiments (for both the GPa3b17 and PMEL TCR systems). This computational approach precisely assesses residue-specific binding energies by substituting target residues with alanine, thereby removing side-chain interactions beyond the β -carbon while preserving the backbone conformation. It enables efficient identification of key interaction hotspots while preserving the backbone structure, offering a rapid, cost-effective tool to guide experimental mutagenesis and rational interface design.^{43,44} First, our FEP calculations of the PMEL17 TCR–pHLA binding complex were in good agreement with experimental measurements (Figure 1C), which confers the applicability of our computational methods to this important antigenic peptide. Concerning GPa3b17 TCR–pHLA binding, Figure 1D demonstrates that the mutations at positions 2, 4, and 9

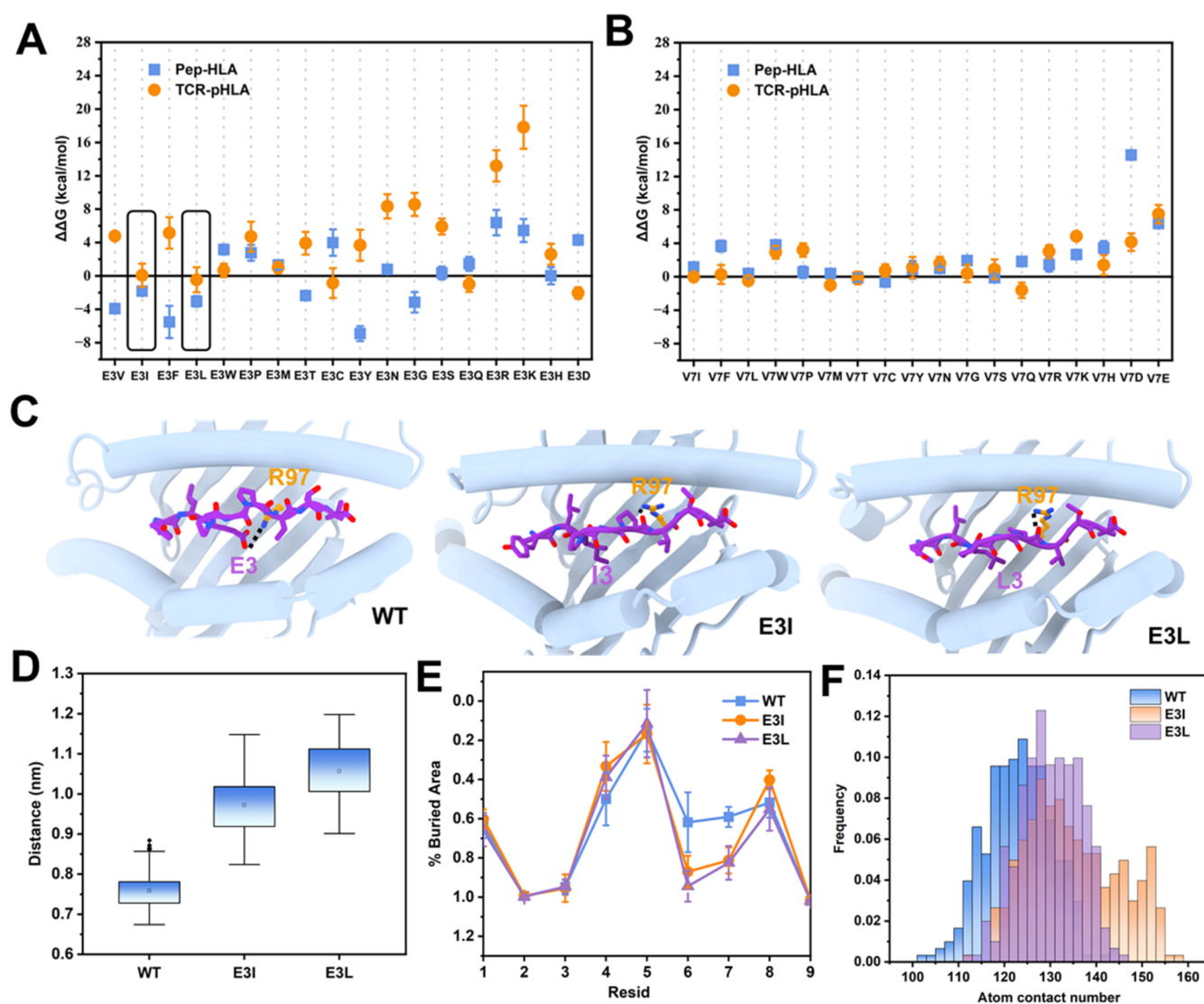


Figure 2. Binding free energy change for peptide–HLA and GPA3b17 TCR–pHLA following exhaustive single mutations at position 3 (A) and position 7 (B) of the peptide. The favorable single mutations are highlighted with black boxes. (C) The binding details of WT, E3I, and E3L peptides with the HLA-A*02:01 molecule. The peptide and R97 residue on the HLA-A*02:01 molecule are shown in purple and orange stick representations, respectively. The HLA-A*02:01 molecule is shown in a blue cartoon representation. For clarity, oxygen and nitrogen atoms are specifically colored red and dark blue, respectively. (D) The distance between the residues at position 3 and the R97 residue on HLA, which is defined as the center of mass (COM) distance between the side chain heavy atoms of the two residues. (E) Buried surface area of the WT, E3I, and E3L peptides in the GPA3b17 TCR–pHLA complex. (F) Distribution of heavy atom contact number between WT, E3I, and E3L peptides and the HLA-A*02:01 molecule.

(L2A, P4A and V9A) significantly weaken the binding between the peptide and the HLA molecule ($\Delta\Delta G = 4.7 \pm 0.3$, 2.2 ± 0.3 , and 2.5 ± 0.3 kcal/mol for L2A, P4A, and V9A, respectively). In contrast, these mutations had negligible effects on the interaction between GPA3b17 TCR and pHLA. The fact that positions 2 and 9 are energetically important for peptide binding to HLA is consistent with their high buried surface area within the HLA molecule (Figure 1B). For mutations P6A and V7A, they showed neutral effects on the binding affinities of both the peptide to HLA and GPA3b17 TCR to pHLA. Notably, mutating the residue at position 5 to alanine largely weakened the binding affinity of GPA3b17 TCR to pHLA ($\Delta\Delta G = 3.4 \pm 0.6$ kcal/mol), without affecting the binding affinity of the peptide to HLA ($\Delta\Delta G = -0.5 \pm 0.3$ kcal/mol). This significant impact on TCR binding affinity suggests the critical role of position 5 in TCR recognition.⁴¹

Interestingly, mutation at position 3 (glutamic acid to alanine) enhanced the binding affinity to HLA ($\Delta\Delta G = -3.1 \pm 0.6$ kcal/mol), but significantly reduced the TCR–pHLA binding ($\Delta\Delta G = 4.6 \pm 0.8$ kcal/mol). Thus, beyond canonical anchor residue preferences at positions 2 and 9, we found that a mutation at the nonanchor position 3 can also make a significant contribution to peptide–HLA binding. Notably, TCR binding is highly sensitive to substitutions at position 3, despite its extensive buried surface area and substantial interactions with the HLA molecule (Figure 1B).

Rational Design of Mimotopes with Enhanced Binding Interactions. Alanine scanning mutagenesis revealed that the GPA3b17 TCR is sensitive to the E3A mutation, with a significant weakening of the binding affinity between TCR and pHLA (Figure 1D). To determine whether other single mutations at position 3 produced a similar response, we

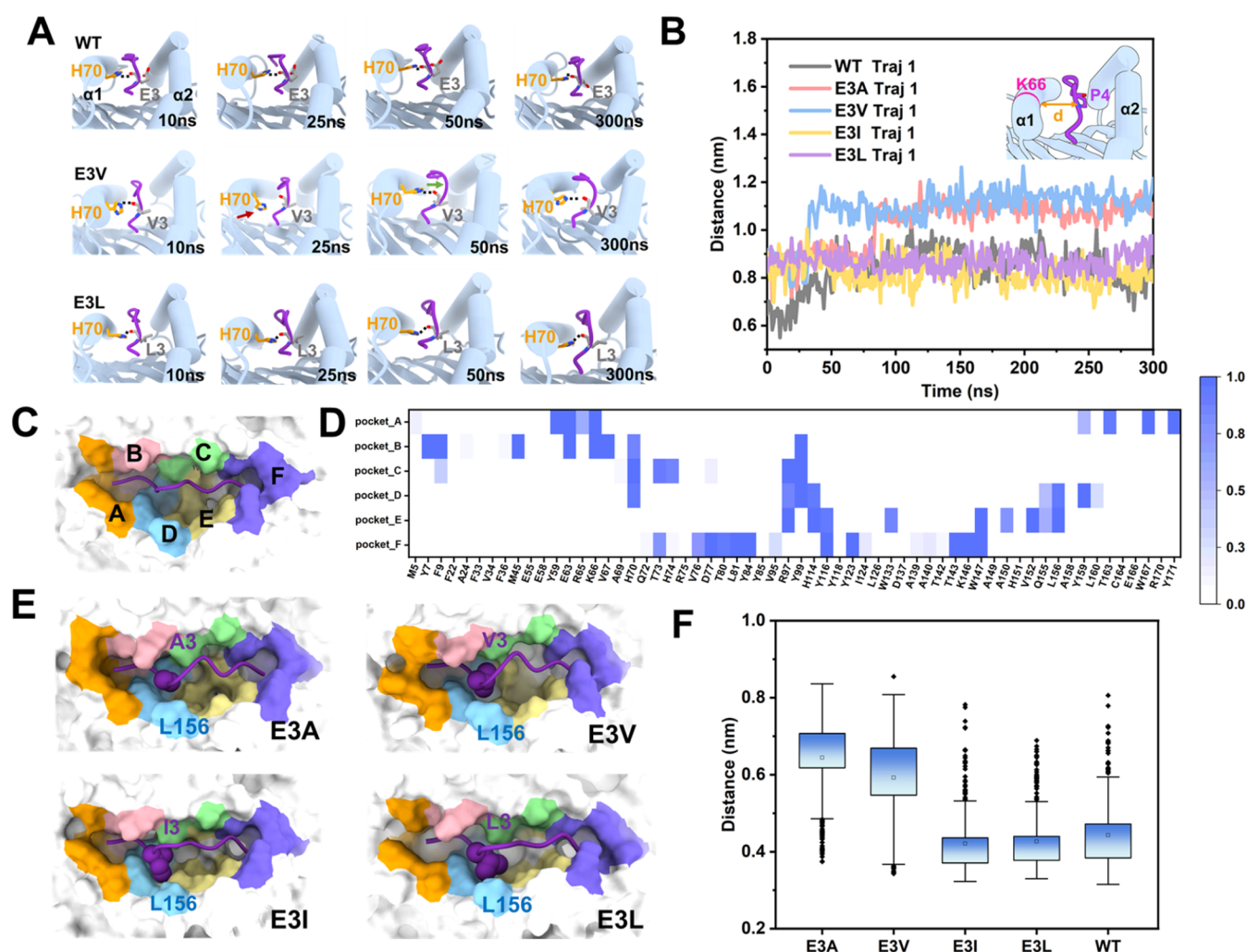


Figure 3. (A) Snapshots showing the binding of WT, E3V, and E3L peptides to the HLA-A*02:01 molecule. The peptide and HLA-A*02:01 molecule are shown in purple and blue cartoon representations, respectively. The E3, V3, and L3 residues on peptides and the H70 residue on the HLA-A*02:01 molecule are shown in gray and orange stick representations, respectively. For clarity, oxygen and nitrogen atoms are specifically shown in red and dark blue, respectively. The red and green arrows highlight the shift in the orientation of the H70 residue and the distortion of the peptide conformation, respectively. (B) The distance between the $\alpha 1$ helix of the HLA-A*02:01 molecule and the torsion portion of the peptide from the representative simulation trajectories. The distance is defined as the distance between the C atom of K66 on the $\alpha 1$ helix and the N atom of P4 on the peptide. (C) Binding pockets A, B, C, D, E, and F of the HLA-A*02:01 molecule. (D) The pocket composition of the HLA-A*02:01 molecule. Color depth represents the probability from 0 to 1. (E) The final conformations of E3A, E3V, E3I, and E3L peptides bound in the HLA-A*02:01 pocket. The residue at position 3 of each peptide is shown in sphere representation. L156 in the D pocket of HLA-A*02:01 is highlighted. (F) Distribution of distances between the residue 3 of E3A, E3V, E3I, and E3L peptides and L156 on HLA-A*02:01. The distance was defined as the COM distance between the terminal heavy atoms of the side chains of residue 3 and L156.

performed exhaustive FEP calculations at this site (Figure 2A). Several mutations at position 3 (E3V, E3F, E3T, E3Y, and E3G) increased the binding affinity between the peptide and HLA, but significantly decreased the binding affinity between GPa3b17 TCR and pHLA. Additionally, some mutations (E3P, E3R, and E3K) reduced both peptide–HLA and TCR–pHLA binding. Notably, the E3I and E3L mutations enhanced the binding affinity of the peptide to HLA ($\Delta\Delta G = -1.8 \pm 0.4$ and -3.0 ± 0.6 kcal/mol for E3I and E3L, respectively), whereas they had little effect on GPa3b17 TCR–pHLA binding ($\Delta\Delta G = 0.1 \pm 1.4$ kcal/mol for E3I, $\Delta\Delta G = -0.5 \pm 1.5$ kcal/mol for E3L). Thus, the E3I and E3L peptides can be treated as potential mimotope candidates for subsequent studies. Moreover, to further validate our FEP calculations, we conducted both forward (wild-type to mutant) and reverse (mutant to wild-type) transformations for several selected

mutations (see Figure S5). The results of these two different transformations were consistent (similar in magnitude but opposite in sign), indicating that our $\Delta\Delta G$ estimates are convergent and robust. Additionally, sequence logo⁴⁵ analysis also revealed the high propensity of residue Ile and Leu at position 3 of the nonamer peptide for HLA-A*02:01 (Figure S6), which also indirectly supports our FEP calculation results here.

Apart from position 3, we also tried to induce mutations at other positions of the peptide to search for potential mimotope candidates. Among them, positions 2 and 9 are critical for anchoring the epitope in the HLA groove,⁴² so mutations at these sites are likely to disrupt peptide–HLA binding. Meanwhile, positions 4 to 6 of the peptide are critical for TCR recognition,⁴¹ which limits their potential for mutational analysis. Consequently, we selected position 7, which exhibits a

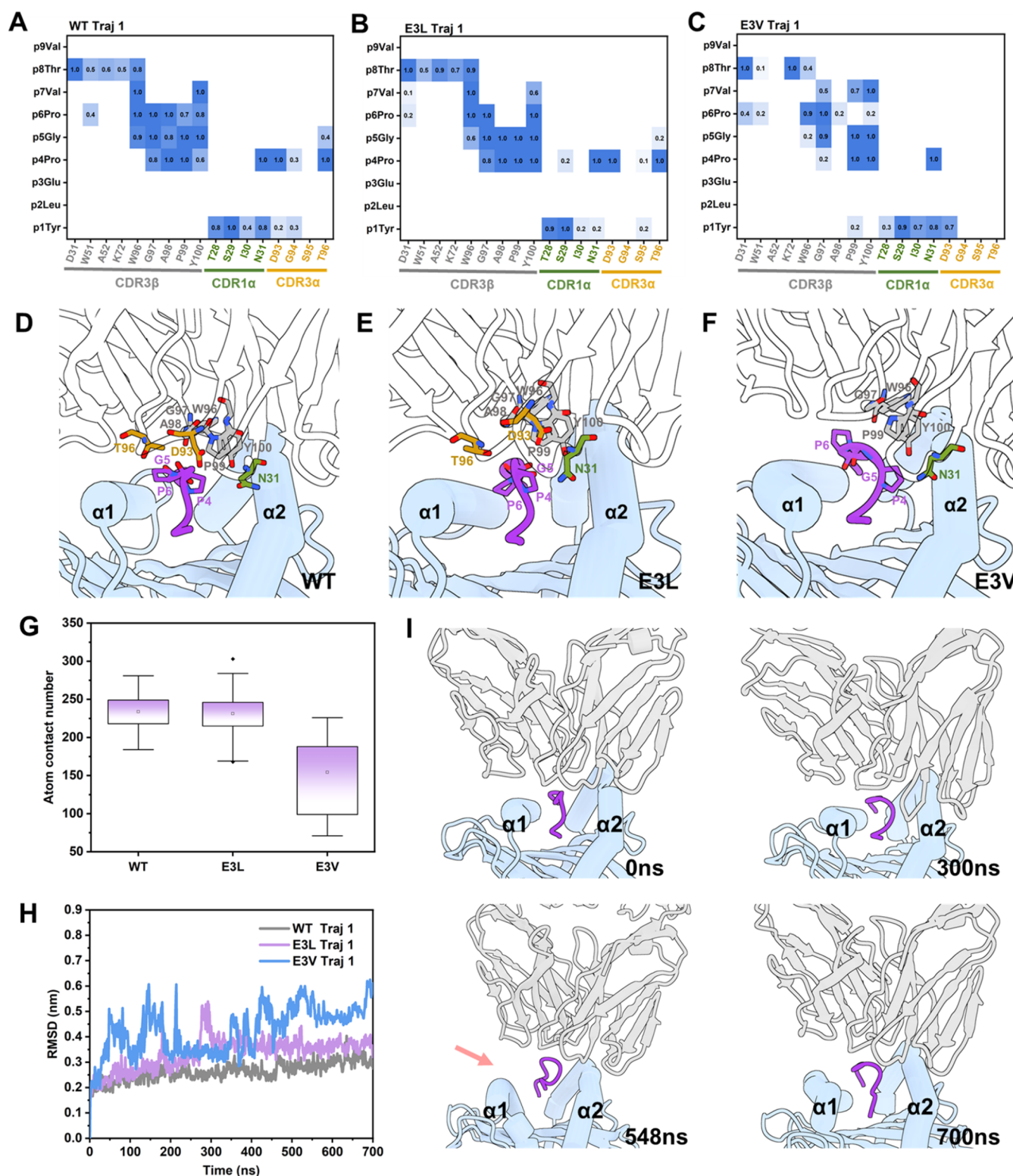


Figure 4. Residue contact probabilities between TCR and the WT peptide (A), E3L peptide (B), and E3V peptide (C) from the representative simulation trajectories over 200–300 ns. Snapshots of key residues (contact probability ≥ 0.5) on GPA3b17 TCR contacting residues 4, 5, and 6 of the WT peptide (D), E3L peptide (E), and E3V peptide (F). The peptide, HLA-A*02:01 and GPA3b17 TCR are shown in purple, blue, and gray cartoon representations, respectively. The residues 4, 5, and 6 on the peptide are shown in purple stick representations, and the key residues on CDR3 β , CDR3 α , and CDR1 α are shown in gray, green, and yellow stick representations, respectively. For clarity, oxygen and nitrogen atoms are specifically shown in red and dark blue, respectively. (G) The atomic contact number between GPA3b17 TCR and pHLA in WT, E3L, and E3V systems (contact is defined as any two heavy atoms within 0.5 nm) averaged over the last 100 ns of all trajectories. (H) The RMSDs as functions of simulation times for the GPA3b17 TCR in WT, E3L, and E3V systems. (I) Snapshots illustrating the binding details of the GPA3b17 TCR-E3V peptide–HLA-A*02:01 complex from the representative simulation trajectory.

considerable buried surface area (Figure 1B), for exhaustive single-residue mutations (Figure 2B). The results showed that most mutations (e.g., V7I, V7L, and V7S) had neutral effects on the binding affinity of peptide–HLA and TCR–pHLA, while the introduction of charged amino acids (V7R, V7K, V7H, V7D, and V7E) significantly weakened both peptide–HLA and TCR–pHLA binding.

To further investigate the effects of E3I and E3L mutations on the peptide–HLA binding, we compared the structures of WT, E3I, and E3L peptides bound to HLA, selecting the final conformations from MD simulations as representatives (Figure 2C). In the WT complex, residue R97 on HLA is stabilized by E3 of the gp100_{280–288(288 V)} peptide through forming a hydrogen bond, as indicated by a center of mass (COM) distance of ~ 0.75 nm between the two residues (Figure 2D). For the E3I and E3L peptides, where Glu3 was mutated to Ile3 or Leu3, R97 on HLA no longer interacts with residue 3. Instead, R97 forms new hydrogen bonds with the main chain at positions 6 and 7, increasing the COM distances to ~ 0.97 nm (I3–R97) and ~ 1.05 nm (L3–R97), respectively. Consequently, the buried surface area of residues at positions 6 and 7 in the E3I and E3L peptides increased compared to the WT peptide, indicating that these positions are pulled toward HLA by more hydrophobic contacts (plus newly formed hydrogen bonds) and thus buried more deeply (Figures 2E and S7). As a result, the E3I and E3L peptides formed more contacts with HLA than the WT peptide (Figure 2F), thereby enhancing the binding affinity between the peptides and HLA (Figure 2A).

Impact of Peptide Conformational Changes on TCR–pHLA Binding Affinity. Although residues Val, Ala, Ile, and Leu all contain hydrophobic side chains, mutations to these hydrophobic residues at epitope position 3 had distinct effects on the TCR–pHLA binding affinity (Figure 2A). To understand this phenomenon, we compared the differences of these peptide variants binding to HLA molecules (Figures 3A and S8). During the first 10 ns of the simulation, H70 on HLA formed stable hydrogen bonds with the backbone of the WT/E3V/E3L peptide, maintaining a consistent distance of ~ 0.85 nm between the $\alpha 1$ helix of HLA and the peptide backbone (Figure 3B). Afterward, for the E3V mutation, H70 flipped downward at $t = 25$ ns, causing a sudden shift in its orientation angle (Figure S9) and breaking the hydrogen bond. After 25 ns, the E3V peptide adopted a twisted conformation, increasing the distance between the $\alpha 1$ helix and the peptide backbone to ~ 1.1 nm (Figure 3B). Then, the E3V peptide maintained this twisted conformation, during which H70 reformed a hydrogen bond with the main chain. This twisted conformation of the E3V peptide is thought to be responsible for the weakened binding affinity between TCR and pHLA, as it may impair TCR recognition (Figure 2A). In contrast, H70 always formed stable hydrogen bonds with the backbones of the WT and E3L peptides (Figure S9). The conformation of the E3L peptide remained nearly unchanged throughout the simulation (similar to that of the WT one), thus having a minimal effect on TCR binding (Figure 2A). Additionally, the E3A peptide exhibited the same twisted conformation as the E3V peptide, while the E3I peptide maintained a stable conformation similar to that of the E3L peptide (Figures 3A and S8). These observations were reproducible in all independent trajectories (Figure S10). Furthermore, we analyzed the dihedral angle distribution of the P4 residue backbone of each peptide to verify these observations (Figure S11). The WT, E3I, and E3L peptides displayed similar

dihedral angle distributions, suggesting that they have similar conformations. In contrast, the dihedral angles of the E3A and E3V peptides were significantly decreased, which is consistent with their twisted conformations. In addition, the profiles of RMSD values of the peptides (Figure S12) and TCR–pHLA complexes (Figure S13) during the simulations also supported the above phenomena; i.e., the twisted conformations of the E3A and E3V mutations have notable destabilizing effects on the structures of the peptide and the ternary complex.

To better understand the molecular mechanism behind the conformational differences among E3A, E3V, E3I, and E3L peptides in the ternary complex, we further investigated the binding details between the position 3 residues and the HLA pocket. Figure 3C illustrates that there are six pockets in the HLA-A*02:01 molecule: A, B, C, D, E, and F.^{46–48} Each contains specific residues (Figure 3D). The residue at position 3 of the peptides primarily interacts with pocket D, which includes amino acids H70, R97, Y99, H114, L156, and Y159,⁴⁹ and their contact details are shown in Figure 3E. Due to the smaller size of A3 and V3, they could not fully occupy the pocket, thus causing potential local instability and leading to twisted peptide conformations (Figures 3A and S8). In contrast, due to the relatively large size of I3 and L3, they were able to fully occupy pocket D like E3 in the WT peptide (Figure S14), thus contributing to a more stable conformation. These points were also confirmed by the distances between the terminal heavy atoms of the side chains of residue 3 in peptides and residue L156 in pocket D. As shown in Figure 3F, the distances between A3–L156 (~ 0.63 nm) and V3–L156 (~ 0.60 nm) are relatively large, indicating a loose contact. In contrast, the distances between I3–L156 (~ 0.40 nm) and L3–L156 ($D = \sim 0.41$ nm) are comparable to those between E3–L156 (~ 0.43 nm), indicating that I3 and L3 form relatively strong interactions with pocket D.

Effects of Different Mutants on TCR Binding. The probability of TCR residues interacting with individual positions of peptides was calculated to reveal the difference in TCR recognition of WT, E3A, E3V, E3I, and E3L peptides (Figure 4A–C and S15). The TCR contact patterns for E3I and E3L peptides were almost identical to those for the WT peptide, while those of the E3A and E3V peptides showed significant differences. In general, positions 4, 5, and 6 of the peptide are crucial for TCR recognition. Therefore, we focused on the TCR residues that interact with these three peptide positions, and their representative conformations are displayed in Figure 4D–F. For WT and E3L peptides, eight residues on the TCR molecule (W96, G97, A98, P99, and Y100 on CDR3 β ; D93 and T96 on CDR3 α ; N31 on CDR1 α , Figure 4A,B,4D,E) formed a stable interaction network with peptide residues 4 to 6. In contrast, the E3V peptide exhibited a twisted conformation that shifted residues 4 and 5 toward the $\alpha 2$ helix of HLA, resulting in the loss of several critical contacts (such as G97, A98, and T96 with position 4, and W96 with position 5, Figure 4C,4F).

To better characterize the impact of different mutants on TCR recognition, we also analyzed the contact numbers between TCR and pHLA (here, a contact was defined as any heavy atom pair with a distance less than 0.5 nm) in the WT, E3L, and E3V systems (Figure 4G). The total contact numbers for the WT and E3L systems were similar, both around 230. However, the contact number for the E3V system was significantly reduced to ~ 160 , which is consistent with the above FEP results (i.e., the binding affinity between TCR and

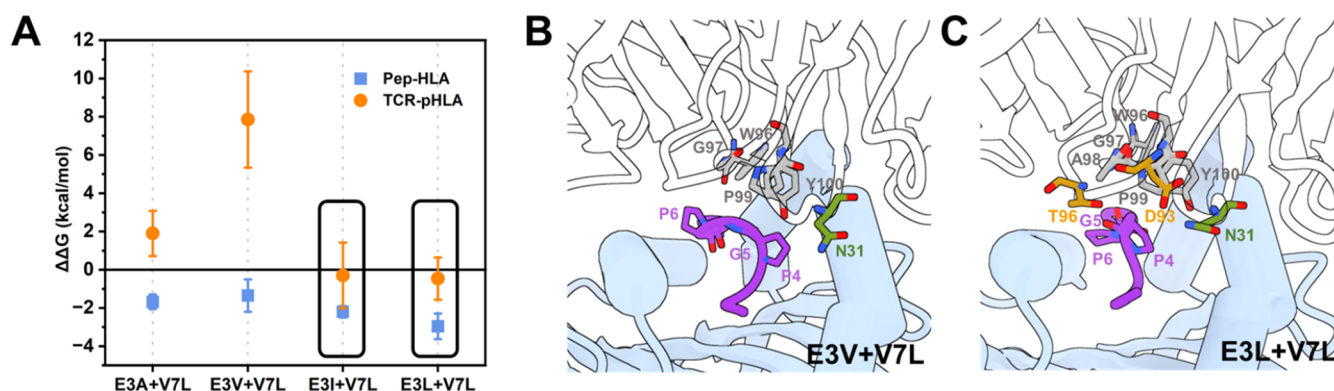


Figure 5. (A) Binding free energy change for peptide–HLA and GPa3b17 TCR–pHLA following double mutation at positions 3 and 7. The favorable double mutations are highlighted with black boxes. Snapshots of key residues (contact probability ≥ 0.5) on GPa3b17 TCR contacting residues at positions 4, 5, and 6 of the E3V+V7L peptide (B) and E3L+V7L peptide (C). The color scheme is consistent with that in Figure 4D–F.

this pHLA was weakened). Additionally, the RMSD of TCR in the E3V system exhibited significant fluctuation compared with those of the WT and E3L systems (Figures 4H and S16A), suggesting that the mutation-induced twisted peptide conformation has an adverse effect on the TCR binding stability. The structural dynamics of the TCR–E3V peptide–HLA ternary complex are shown in Figure 4I. At 300 ns, the peptide exhibited a twisted conformation, which could destabilize TCR–pHLA binding. At $t = 548$ ns, the TCR tilted toward the $\alpha 2$ helix of HLA, separating it from the $\alpha 1$ helix and reducing the contact number between the TCR and $\alpha 1$ helix (Figure S16B). Meanwhile, the peptide conformation shifted toward another structure with a sharp increase in the RMSD (Figure S16C). Until $t = 700$ ns, the TCR remained separated from the $\alpha 1$ helix of HLA. Therefore, although a single mutation in the peptide may enhance its binding affinity for HLA (Figure 2A), it may induce significant conformational changes in the peptide, leading to the destabilization of TCR–pHLA binding. This highlights the importance of considering mutation-induced conformational changes when designing optimized mimotopes.

Combinatorial Mutations at Positions 3 and 7 on Peptide. We also adopted a double-mutation strategy (integrating mutations at positions 3 and 7) to search for potential mimotopes, and the FEP calculation results are shown in Figure 5A. The E3I+V7L and E3L+V7L mutations significantly enhanced the binding affinity between the peptide and HLA ($\Delta\Delta G = -2.2 \pm 0.3$ and -3.0 ± 0.7 kcal/mol, respectively) without weakening the interaction between TCR and pHLA, making them potential candidates for mimotopes. Conversely, the E3V+V7L and E3A+V7L mutations considerably weakened the binding of TCR to pHLA. The trends in free energy changes for the double mutations are consistent with the corresponding single mutations at position 3, indicating that their primary contributions to the free energy changes came from position 3. Figure 5B–C shows the binding conformations of the E3V+V7L and E3L+V7L double mutants, which are almost identical to the binding conformations of the E3V and E3L mutants (Figure 3A). Meanwhile, E3A+V7L and E3I+V7L mutants displayed TCR contact patterns similar to those of the E3A and E3I mutants (Figures S15 and S17). Overall, these four double mutations exhibited structural features similar to the corresponding single mutations at position 3, suggesting the dominant role of the mutations at the key position 3.

CONCLUSIONS AND DISCUSSION

In this work, we conducted extensive computational studies to investigate the binding characteristics of the TCR–pHLA complex related to the gp100_{280–288(288 V)} epitope, utilizing the latest affinity-matured TCR- $\alpha\beta$ (GPa3b17) structure, and designed a series of potential mimotopes. Our free energy perturbation (FEP) results demonstrated that E3I (YLIPGPVTV) and E3L (YLLPGPVTV) peptide mutations can enhance their binding affinity to HLA and have negligible effects on TCR binding, indicating their potential as neo-epitope candidates. However, we also found that some mutations at position 3, such as E3V (YLVPGPVTV) and E3A (YLAPGPVTV), can enhance their binding ability to HLA but significantly weaken the binding affinity of TCR to pHLA. Further MD simulations showed that incomplete occupancy of the HLA pocket by small-sized residues Ala and Val can lead to a twisted peptide conformation, which may interfere with the binding ability of pHLA to TCR. In addition, the combined double mutations E3I+V7L (YLIPGPIVTV) and E3L+V7L (YLLPGPIVTV) exhibit binding properties similar to those of the E3I and E3L mutations, suggesting that they could also serve as promising antigenic mimotopes. Notably, these findings extend beyond the well-established role of anchor residues at positions 2 and 9, demonstrating that mutations at the nonanchor position 3 (specifically E3I and E3L) can enhance peptide–HLA interactions. However, in addition to their enhancing effects on peptide–HLA binding, some mutations (e.g., E3V and E3A) may induce a twisted peptide conformation that significantly impairs TCR recognition, highlighting the critical role of peptide conformation in modulating TCR engagement. Therefore, our results suggest that the effect of a single mutation on peptide structural stability must be considered when designing optimized mimotopes.

Our *in silico* mutagenesis studies provide valuable insights into the effects of mutations in the gp100 mimotope on structural properties and binding capacity, enabling rapid and systematic evaluation of peptide–HLA and TCR–pHLA interactions in various peptide mutants. This computational analysis offers a cost-effective, high-throughput approach to screening and optimizing vaccine candidates. However, we acknowledge that our computational approach still has many limitations in predicting *in vivo* outcomes, as it cannot fully capture the complex immunogenicity processes or stability of peptides under physiological environments. Therefore, exper-

imental validation will be essential for the future confirmation and extension of our proposed computational findings here. For example, binding affinity can be quantified by surface plasmon resonance (SPR), while cellular assays such as IFN- γ ELISPOT and cytotoxicity tests can assess CD8⁺ T-cell activation. In vivo studies using HLA-A*02:01 transgenic mice can further evaluate the immunogenicity, stability, and antitumor efficacy of the designed mimotopes. Additionally, serum stability and protease resistance assays are critical for assessing the durability of the peptides. Combining these experimental measurements with computational predictions and performing multiple rounds of iterations will enhance the reliability and applicability of the proposed mimotopes.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.5c01507>.

Summary of simulation systems; crystal structure of the PMEL17 TCR–pHLA complex; thermodynamic perturbation cycle; RMSDs of the GP3b17 TCR–pHLA complex; forward and reverse free energy changes; sequence logo of HLA-A*02:01 peptide contact sites; structural alignment of peptide–HLA complexes; simulation snapshots showing E3A and E3I peptide binding; orientation angle changes of the H70 residue in HLA; distance between the α 1 helix of HLA and the peptide torsion region; dihedral angle distributions of different mutants; RMSDs of various peptides and TCR–pHLA complexes; final conformations of the WT peptide in the HLA pocket; residue contact probabilities between TCR and single/double mutants; RMSDs of the GP3b17 TCR; atom contact numbers between the WT, E3L, and E3V peptides and the α 1 helix of HLA (PDF)

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Author Contributions

[†]R.Y. and Y.Y. contributed equally to this work. R.Z. conceived the idea and designed the research. R.Y. and Y.Y. carried out molecular dynamics simulations and free energy perturbation calculations. R.Y., Y.Y., K.C.C., D.Z. and R.Z. analyzed the data and wrote the manuscript. All authors participated in discussions and revisions of the manuscript.

Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

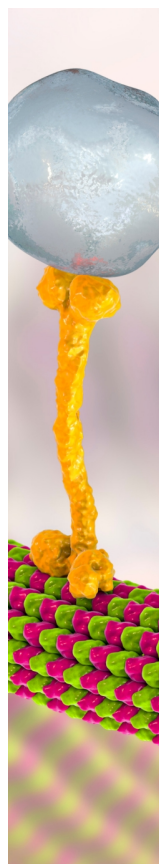
FEPfree-energy perturbation; MDmolecular dynamics; CTLcytotoxic T lymphocyte; TCRT-cell receptor; HLAhuman leukocyte antigen; gp100glycoprotein 100; WTwild type; RMSDroot-mean-square deviation; COMcenter of mass

■ REFERENCES

- (1) Salgaller, M. L.; Marincola, F. M.; Cormier, J. N.; Rosenberg, S. A. Immunization against Epitopes in the Human Melanoma Antigen Gp100 Following Patient Immunization with Synthetic Peptides. *Cancer Res.* **1996**, *56*, 4749–4757.
- (2) Boon, T.; Coulie, P. G.; Eynde, B. J. V. d.; Bruggen, P. v. d. Human T Cell Responses against Melanoma. *Annu. Rev. Immunol.* **2006**, *24*, 175–208.
- (3) Kalaora, S.; Nagler, A.; Wargo, J. A.; Samuels, Y. Mechanisms of Immune Activation and Regulation: Lessons from Melanoma. *Nat. Rev. Cancer* **2022**, *22*, 195–207.
- (4) Martinez-Perez, D.; Viñal, D.; Solares, I.; Espinosa, E.; Feliu, J. Gp-100 as a Novel Therapeutic Target in Uveal Melanoma. *Cancers* **2021**, *13*, 5968.
- (5) Raposo, G.; Marks, M. S. Melanosomes—Dark Organelles Enlighten Endosomal Membrane Transport. *Nat. Rev. Mol. Cell Biol.* **2007**, *8*, 786–797.
- (6) Wagner, S. N.; Wagner, C.; Schultewolter, T.; Goos, M. Analysis of Pmel17/Gp100 Expression in Primary Human Tissue Specimens: Implications for Melanoma Immuno- and Gene-Therapy. *Cancer Immunol., Immunother.* **1997**, *44*, 239–247.
- (7) Centeno, P. P.; Pavet, V.; Marais, R. The Journey from Melanocytes to Melanoma. *Nat. Rev. Cancer* **2023**, *23*, 372–390.
- (8) Rosenberg, S. A.; Yang, J. C.; Schwartzentruber, D. J.; et al. Immunologic and Therapeutic Evaluation of a Synthetic Peptide Vaccine for the Treatment of Patients with Metastatic Melanoma. *Nat. Med.* **1998**, *4*, 321–327.

- (9) Kawakami, Y.; Eliyahu, S.; Jennings, C.; Sakaguchi, K.; Kang, X.; Southwood, S.; Robbins, P. F.; Sette, A.; Appella, E.; Rosenberg, S. A. Recognition of Multiple Epitopes in the Human Melanoma Antigen Gp100 by Tumor-Infiltrating T Lymphocytes Associated with in Vivo Tumor Regression. *J. Immunol.* **1995**, *154*, 3961–3968.
- (10) Kawakami, Y.; Eliyahu, S.; Sakaguchi, K.; Robbins, P. F.; Rivoltini, L.; Yannelli, J. R.; Appella, E.; Rosenberg, S. A. Identification of the Immunodominant Peptides of the Mart-1 Human Melanoma Antigen Recognized by the Majority of Hla-A2-Restricted Tumor Infiltrating Lymphocytes. *J. Exp. Med.* **1994**, *180*, 347–352.
- (11) Kawakami, Y.; Eliyahu, S.; Delgado, C. H.; Robbins, P. F.; Sakaguchi, K.; Appella, E.; Yannelli, J. R.; Adema, G. J.; Miki, T.; Rosenberg, S. A. Identification of a Human Melanoma Antigen Recognized by Tumor-Infiltrating Lymphocytes Associated with in Vivo Tumor Rejection. *Proc. Natl. Acad. Sci. U.S.A.* **1994**, *91*, 6458–6462.
- (12) Parkhurst, M. R.; Salgaller, M. L.; Southwood, S.; Robbins, P. F.; Sette, A.; Rosenberg, S. A.; Kawakami, Y. Improved Induction of Melanoma-Reactive Ctl with Peptides from the Melanoma Antigen Gp100 Modified at Hla-a* 0201-Binding Residues. *J. Immunol.* **1996**, *157*, 2539–2548.
- (13) Cox, A. L.; Skipper, J.; Chen, Y.; Henderson, R. A.; Darrow, T. L.; Shabanowitz, J.; Engelhard, V. H.; Hunt, D. F.; Slingluff, C. L., Jr. Identification of a Peptide Recognized by Five Melanoma-Specific Human Cytotoxic T Cell Lines. *Science* **1994**, *264*, 716–719.
- (14) Pereira, A. C. L.; Bezerra, K.; Santos, J.; Oliveira, J. I.; Freire, V.; Fulco, U. In Silico Approach of Modified Melanoma Peptides and Their Immunotherapeutic Potential. *Phys. Chem. Chem. Phys.* **2021**, *23*, 2836–2845.
- (15) Schaft, N.; Coccors, M.; Drexhage, J.; Knoop, C.; de Vries, I. J. M.; Adema, G. J.; Debets, R. An Altered Gp100 Peptide Ligand with Decreased Binding by Tcr and Cd8 α Dissects T Cell Cytotoxicity from Production of Cytokines and Activation of Nfat. *Front. Immunol.* **2013**, *4*, No. 270.
- (16) Meloen, R.; Puijk, W.; Slootstra, J. Mimotopes: Realization of an Unlikely Concept. *J. Mol. Recognit.* **2000**, *13*, 352–359.
- (17) Miles, K. M.; Miles, J. J.; Madura, F.; Sewell, A. K.; Cole, D. K. Real Time Detection of Peptide–Mhc Dissociation Reveals That Improvement of Primary Mhc-Binding Residues Can Have a Minimal, or No, Effect on Stability. *Mol. Immunol.* **2011**, *48*, 728–732.
- (18) Lesterhuis, W. J.; Schreiber, G.; Scharenborg, N. M.; Brouwer, H. M.-I. H.; Gerritsen, M.-J. P.; Croockewit, S.; Coulie, P. G.; Torensma, R.; Adema, G. J.; Figdor, C. G.; et al. Wild-Type and Modified Gp100 Peptide-Pulsed Dendritic Cell Vaccination of Advanced Melanoma Patients Can Lead to Long-Term Clinical Responses Independent of the Peptide Used. *Cancer Immunol., Immunother.* **2011**, *60*, 249–260.
- (19) Sušac, L.; Vuong, M. T.; Thomas, C.; von Bülow, S.; O'Brien-Ball, C.; Santos, A. M.; Fernandes, R. A.; Hummer, G.; Tampé, R.; Davis, S. J. Structure of a Fully Assembled Tumor-Specific T Cell Receptor Ligated by Pmhc. *Cell* **2022**, *185*, 3201–3213.
- (20) Bianchi, V.; Bulek, A.; Fuller, A.; Lloyd, A.; Attaf, M.; Rizkallah, P. J.; Dolton, G.; Sewell, A. K.; Cole, D. K. A Molecular Switch Abrogates Glycoprotein 100 (Gp100) T-Cell Receptor (Tcr) Targeting of a Human Melanoma Antigen. *J. Biol. Chem.* **2016**, *291*, 8951–8959.
- (21) Phillips, J. C.; Braun, R.; Wang, W.; Gumbart, J.; Tajkhorshid, E.; Villa, E.; Chipot, C.; Skeel, R. D.; Kale, L.; Schulten, K. Scalable Molecular Dynamics with Namd. *J. Comput. Chem.* **2005**, *26*, 1781–1802.
- (22) Brooks, B. R.; Brucoleri, R. E.; Olafson, B. D.; States, D. J.; Swaminathan, S.; Karplus, M. Charmm - a Program for Macromolecular Energy, Minimization, and Dynamics Calculations. *J. Comput. Chem.* **1983**, *4*, 187–217.
- (23) Klauda, J. B.; Venable, R. M.; Freites, J. A.; O'Connor, J. W.; Tobias, D. J.; Mondragon-Ramirez, C.; Vorobyov, I.; MacKerell, A. D.; Pastor, R. W. Update of the Charmm All-Atom Additive Force Field for Lipids: Validation on Six Lipid Types. *J. Phys. Chem. B* **2010**, *114*, 7830–7843.
- (24) Mackerell, A. D.; Feig, M.; Brooks, C. L. Extending the Treatment of Backbone Energetics in Protein Force Fields: Limitations of Gas-Phase Quantum Mechanics in Reproducing Protein Conformational Distributions in Molecular Dynamics Simulations. *J. Comput. Chem.* **2004**, *25*, 1400–1415.
- (25) Huang, J.; MacKerell, A. D., Jr. Charmm36 All-Atom Additive Protein Force Field: Validation Based on Comparison to Nmr Data. *J. Comput. Chem.* **2013**, *34*, 2135–2145.
- (26) Lee, J.; Cheng, X.; Jo, S.; MacKerell, A. D.; Klauda, J. B.; Im, W. Charmm-Gui Input Generator for Namd, Gromacs, Amber, Openmm, and Charmm/Openmm Simulations Using the Charmm36 Additive Force Field. *Biophys. J.* **2016**, *110*, No. 641a.
- (27) Jorgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. L. Comparison of Simple Potential Functions for Simulating Liquid Water. *J. Chem. Phys.* **1983**, *79*, 926–935.
- (28) Darden, T.; York, D.; Pedersen, L. Particle Mesh Ewald - an N.Log(N) Method for Ewald Sums in Large Systems. *J. Chem. Phys.* **1993**, *98*, 10089–10092.
- (29) Bussi, G.; Donadio, D.; Parrinello, M. Canonical Sampling through Velocity Rescaling. *J. Chem. Phys.* **2007**, *126*, No. 014101.
- (30) Parrinello, M.; Rahman, A. Polymorphic Transitions in Single Crystals a New Molecular Dynamics Method. *J. Appl. Phys.* **1981**, *52*, 7182–7190.
- (31) Hess, B.; Bekker, H.; Berendsen, H. J. C.; Fraaije, J. G. E. M. Lincs: A Linear Constraint Solver for Molecular Simulations. *J. Comput. Chem.* **1997**, *18*, 1463–1472.
- (32) Pettersen, E. F.; Goddard, T. D.; Huang, C. C.; Couch, G. S.; Greenblatt, D. M.; Meng, E. C.; Ferrin, T. E. Ucsf Chimera—a Visualization System for Exploratory Research and Analysis. *J. Comput. Chem.* **2004**, *25*, 1605–1612.
- (33) Ahmed, R.; Omidian, Z.; Giwa, A.; Cornwell, B.; Majety, N.; Bell, D. R.; Lee, S.; Zhang, H.; Michels, A.; Desiderio, S.; et al. A Public Bcr Present in a Unique Dual-Receptor-Expressing Lymphocyte from Type 1 Diabetes Patients Encodes a Potent T Cell Autoantigen. *Cell* **2019**, *177*, 1583–1599.
- (34) Chowell, D.; Morris, L. G. T.; Grigg, C. M.; et al. Patient Hla Class I Genotype Influences Cancer Response to Checkpoint Blockade Immunotherapy. *Science* **2018**, *359*, 582–587.
- (35) Joglekar, A. V.; Liu, Z.; Weber, J. K.; et al. T Cell Receptors for the Hiv Kk10 Epitope from Patients with Differential Immunologic Control Are Functionally Indistinguishable. *Proc. Natl. Acad. Sci. U.S.A.* **2018**, *115*, 1877–1882.
- (36) Lima Silva, W. J.; Ferreira de Freitas, R. Assessing the Performance of Docking, Fep, and Mm/Gbsa Methods on a Series of Klk6 Inhibitors. *J. Comput.-Aided Mol. Des.* **2023**, *37*, 407–418.
- (37) Song, Y.; Bell, D. R.; Ahmed, R.; Chan, K. C.; Lee, S.; Hamad, A. R. A.; Zhou, R. A Mutagenesis Study of Autoantigen Optimization for Potential T1d Vaccine Design. *Proc. Natl. Acad. Sci. U.S.A.* **2023**, *120*, No. e2214430120.
- (38) Zhang, D.; Qiao, L.; Lei, X.; Dong, X.; Tong, Y.; Wang, J.; Wang, Z.; Zhou, R. Mutagenesis and Structural Studies Reveal the Basis for the Specific Binding of Sars-Cov-2 Sl3 Rna Element with Human Tia1 Protein. *Nat. Commun.* **2023**, *14*, No. 3715.
- (39) Ngo, S. T.; Tam, N. M.; Pham, M. Q.; Nguyen, T. H. Benchmark of Popular Free Energy Approaches Revealing the Inhibitors Binding to Sars-Cov-2 Mpro. *J. Chem. Inf. Model.* **2021**, *61*, 2302–2312.
- (40) Klimovich, P. V.; Shirts, M. R.; Mobley, D. L. Guidelines for the Analysis of Free Energy Calculations. *J. Comput.-Aided Mol. Des.* **2015**, *29*, 397–411.
- (41) Matsumura, M.; Fremont, D. H.; Peterson, P. A.; Wilson, I. A. Emerging Principles for the Recognition of Peptide Antigens by Mhc Class I Molecules. *Science* **1992**, *257*, 927–934.
- (42) Barouch, D.; Friede, T.; Stevanović, S.; Tussey, L.; Smith, K.; Rowland-Jones, S.; Braud, V.; McMichael, A.; Rammensee, H.-G. Hla-A2 Subtypes Are Functionally Distinct in Peptide Binding and Presentation. *J. Exp. Med.* **1995**, *182*, 1847–1856.

- (43) Massova, I.; Kollman, P. A. Computational Alanine Scanning to Probe Protein–Protein Interactions: A Novel Approach to Evaluate Binding Free Energies. *J. Am. Chem. Soc.* **1999**, *121*, 8133–8143.
- (44) Pál, G.; Ultsch, M. H.; Clark, K. P.; Currell, B.; Kossiakoff, A. A.; Sidhu, S. S. Intramolecular Cooperativity in a Protein Binding Site Assessed by Combinatorial Shotgun Scanning Mutagenesis. *J. Mol. Biol.* **2005**, *347*, 489–494.
- (45) Thomsen, M. C. F.; Nielsen, M. Seq2logo: A Method for Construction and Visualization of Amino Acid Binding Motifs and Sequence Profiles Including Sequence Weighting, Pseudo Counts and Two-Sided Representation of Amino Acid Enrichment and Depletion. *Nucleic Acids Res.* **2012**, *40*, W281–W287.
- (46) Falk, K.; Rotzschke, O.; Stevanovic, S.; Jung, G.; Rammensee, H. G. Allele-Specific Motifs Revealed by Sequencing of Self-Peptides Eluted from Mhc Molecules. *Nature* **1991**, *351*, 290–296.
- (47) Feng, M.; Chan, K. C.; Zhong, Q.; Zhou, R. In Silico Design of High-Affinity Antigenic Peptides for Hla-B44. *Int. J. Biol. Macromol.* **2024**, *267*, No. 131356.
- (48) Hunt, D. F.; Henderson, R. A.; Shabanowitz, J.; Sakaguchi, K.; Michel, H.; Sevilir, N.; Cox, A. L.; Appella, E.; Engelhard, V. H. Characterization of Peptides Bound to the Class I Mhc Molecule Hla-A2.1 by Mass Spectrometry. *Science* **1992**, *255*, 1261–1263.
- (49) Nguyen, A. T.; Szeto, C.; Gras, S. The Pockets Guide to Hla Class I Molecules. *Biochem. Soc. Trans.* **2021**, *49*, 2319–2331.



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